HDOC-67

Recursive GCD Using Euclid's Algorithm

Question

Write a recursive function to find the GCD of two positive integers using Euclid's algorithm.

Algorithm / Logic

1. Start the program.
2. Read two positive integers a and b from the user.
3. Call the recursive function gcd(a, b).
4. If b is equal to 0, return a. This is the base case.
5. Otherwise, recursively call gcd(b, a % b).
6. Repeat the recursive process until the second number becomes 0.
7. Display the returned value as the GCD of the two numbers.
8. Stop the program.

C Program

```
#include <stdio.h>

int gcd(int a, int b)
{
    if (b == 0)
        return a;

    return gcd(b, a % b);
}

int main()
{
    int num1, num2;

    printf("Enter two positive integers: ");
    scanf("%d %d", &num1, &num2);

    printf("GCD of %d and %d = %d\n",
           num1, num2, gcd(num1, num2));

    return 0;
}
```

Explanation

The program uses Euclid's algorithm to calculate the Greatest Common Divisor (GCD). The recursive function gcd(a, b) checks whether b is 0. If b is 0, a is returned as the GCD. Otherwise, the function calls itself with b and the remainder of a divided by b. This continues until the remainder becomes 0.

For example, for 48 and 18: gcd(48,18) -> gcd(18,12) -> gcd(12,6) -> gcd(6,0). Therefore, the GCD is 6.

Test Cases

Test Case	First Number	Second Number	Expected Output	Actual Output
1	48	18	6	6
2	100	25	25	25
3	17	13	1	1
4	54	24	6	6
5	81	27	27	27

Sample Output

```
Enter two positive integers: 48 18
GCD of 48 and 18 = 6
```

Result

Thus, the C program successfully finds the GCD of two positive integers using a recursive function based on Euclid's algorithm.